Investment in Storage and Hydrogen

Group 15 | Technical report, Tier 1 | 2026/27

Executive summary

Under economic dispatch in 2023, one 20 MW electrolyser and no battery add £8.57m per year at a netback of £7.799/kg and an offtake cap of 10 t/day. Daily electricity priority selects zero investment. The 20 MW plant captures 13.92 GWh of 549.94 GWh of physical surplus. The rest of its electricity input, 157.92 GWh, would otherwise have been sold. The economic case rests on the value of that diversion and the buyer's terms.

1 Decision context and data

The constructed system combines the supplied generation portfolio with the simulated wind farm [1]. Existing generation is fixed. Demand caps electricity sales, imports are excluded and unused output has zero value. Daily mean MW is multiplied by 24 h. We use 2023 to choose capacity.

Table 1. Input checks and repairs. The audit records replacements and flags estimates. Date counts can overlap [2].

Check	Treatment
Calendar and totals	Align five series to 2,452 UTC days (January 2020 to 17 September 2026). Reconcile generation by fuel to total generation.
Midnight snapshots	Rebuild full days for 7 generation, 350 demand and 346 price endpoints.
Price aggregation	Volume-weight active providers within periods, then average across the day. Rebase to 2020 pounds with monthly CPI [3].
Gaps and anomalous zeros	Use observed intervals where available, then flag interpolated or proxy values. A strict-quality sensitivity retains 280 days in 2023.

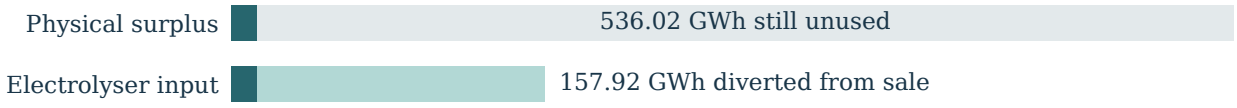


Figure 1. 2023 physical surplus and electrolyser input under A. The dark segment in each bar is the same 13.92 GWh of captured surplus.

2 Tier 1. Flexibility investment

2.1 Technology and economic assumptions

The brief's EUR 288,000/MWh converts to £256,235/MWh at the 2020 exchange rate [1, 8]. For its £9.50/kg support benchmark, we use the published 2012 basis: £175/MWh times 39.4 kWh/kg, then rebase by CPI to £7.799/kg [7, 3, 8]. The 10 t/day offtake cap is a buyer scenario.

Table 2. Equipment and finance assumptions. All model prices are in 2020 pounds.

Input	Assumption
Battery	1.5 MW / 3 MWh per module. 20–80% state of charge, 90% round-trip efficiency, £14,862/MW-year fixed cost, 15-year life [1, 4].
Electrolyser	20 MW per module, 20–100% online load and 55.9 kWh/kg. Installed cost £29.21m per module (IEA midpoint). UK cost proxies give £0.517m fixed cost per year and £9.79m renewals in years 11 and 22 of a 30-year life [5, 6, 8].
Finance	5% real discount rate. Hydrogen netback is after non-power variable costs.

2.2 Mixed-integer formulation

For days $t \in \mathcal{T}$, supply G_t and demand D_t are in MWh. The wholesale price p_t is in £/MWh, hydrogen netback v is in £/kg, and L limits hydrogen sales in kg/day. The decision variables q_t, c_t, r_t, h_t, u_t are direct sale, charge, discharge, electrolyser input and unused generation in MWh. Installed module counts are b, n for the battery and electrolyser, and z_t counts online electrolyser modules. The model uses $\kappa = 0.0559$ MWh/kg and equivalent annual costs a_B, a_H. The sample represents Y years, with $Y = 1$ for a full calendar year and $Y = 280/365$ in the strict-quality case.

$$\max \quad \frac{1}{Y}\sum_{t\in\mathcal{T}}\left[p_t(q_t + r_t) + v\frac{h_t}{\kappa}\right] - a_B b - a_H n, \tag{1a}$$

$$\text{subject to} \quad q_t + c_t + h_t + u_t = G_t, \qquad q_t + r_t \le D_t, \tag{1b}$$

$$c_t \le E_B b/\eta, \qquad r_t \le \eta E_B b, \qquad \sum_{t\in\mathcal{T}} r_t = \eta^2 \sum_{t\in\mathcal{T}} c_t, \tag{1c}$$

$$E_H^{\min} z_t \le h_t \le E_H^{\max} z_t, \qquad h_t/\kappa \le L, \qquad z_t \le n, \tag{1d}$$

$$q_t, c_t, r_t, h_t, u_t \ge 0, \qquad b, n, z_t \in \mathbb{Z}_{\ge 0}. \tag{1e}$$

The day-indexed constraints apply for all t. Equation (1b) allocates supply and limits electricity sales. Equation (1c) allows one usable battery window per day and balances energy over the year, with $E_B = 1.8$ MWh and $\eta = \sqrt{0.90}$. In (1d), $E_H^{\min} = 96$ and $E_H^{\max} = 480$ MWh bound daily input per online module and L caps output. Policy A uses economic dispatch. Policy B additionally fixes $q_t = \min\{G_t, D_t\}$. B therefore incurs negative sales revenue when the price is negative (two days in 2020, none in 2023).

For technology $j \in \{B, H\}$, equivalent annual cost is

$$a_j = F_j + \frac{i}{1-(1+i)^{-N_j}}\left(K_j + \sum_{k\in\mathcal{R}_j}\frac{C_{jk}}{(1+i)^k}\right), \tag{2}$$

where i is the real discount rate, N_j the service life, F_j annual fixed cost, K_j upfront cost including commissioning and C_{jk} the cost of a renewal in year $k \in \mathcal{R}_j$. Gains subtract the zero-investment objective under the same policy.

2.3 Portfolio results and matched operating rule

The simple rule sells electricity first, then produces hydrogen from the remainder subject to loading and offtake limits. With 20 MW it yields −£1.07m/year. Economic dispatch adds £9.64m/year over the rule at the same capacity. Both capture 13.92 GWh of otherwise-unused power. The additional value in A comes from redirecting direct sales.

Table 3. 2023 portfolio comparison. Capacity is battery / electrolyser (MW). Gains are versus zero investment under the same policy.

	A. Economic dispatch		B. Daily electricity priority	
Scenario	Capacity	Gain (£m/yr)	Capacity	Gain (£m/yr)
No investment	0 / 0	0.00	0 / 0	0.00
Battery only	0 / 0	0.00	0 / 0	0.00
Electrolyser only	0 / 20	8.57	0 / 0	0.00
Joint capacity choice	0 / 20	8.57	0 / 0	0.00
Fixed 20 MW module	0 / 20	8.57	0 / 20	−1.07

In A, the 20 MW investment reduces annual electricity sales from 215,102.16 to 214,944.24 GWh. Demand coverage falls from 87.512% to 87.448%.

2.4 Energy capture and capacity limits

Physical surplus is $S_t = \max\{G_t - D_t, 0\}$. In 2023, A's zero-investment case leaves 549.94 GWh unused; the 20 MW case leaves 536.02 GWh. The module runs at 0.48 GWh on all 29 surplus days. It captures 13.92 GWh, or 2.53% of annual surplus and 8.1% of its input (Figure 1). At 10 t/day, the buyer can absorb hydrogen made from at most 0.559 GWh/day, so even with larger equipment this contract caps pure surplus capture at 16.21 GWh in 2023.

For A with no battery, attribute $C_t = \min\{h_t, S_t\}$ to surplus first. With wind and solar generation R_t, their share W_t of that energy obeys

$$\max\{0, C_t - (G_t - R_t)\} \le W_t \le \min\{C_t, R_t\}. \tag{3}$$

The annual range is 0–13.92 GWh, also when biomass and run-of-river hydro are included. Source tracing is required to identify the captured fuel mix.

2.5 Contract terms and service requirements

At 10 t/day, the fixed 20 MW module breaks even at £4.61/kg in A and £12.08/kg in B. A second module can sell more hydrogen under that cap, but annual gain falls to £7.47m from £8.57m. With 50 t/day of acceptance, A selects 120 MW with £49.09m/year of gain on £175.25m of upfront capital. Figure 2 shows the lower offtake range in both policies.

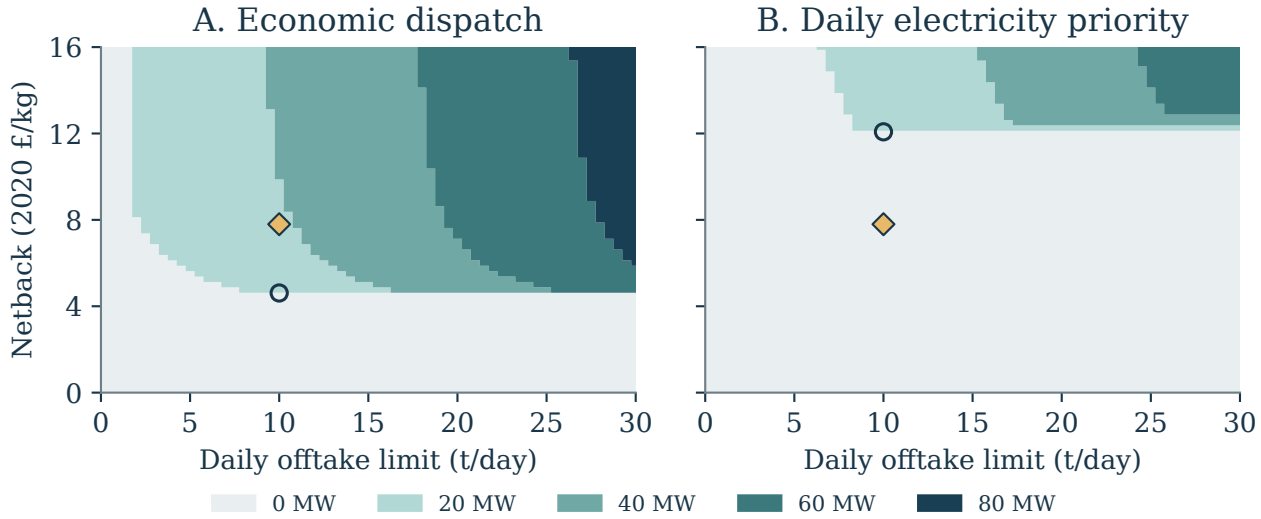


Figure 2. 2023 optimal capacity over the contract grid. Diamonds show the reference terms. Open circles show the fixed 20 MW break-even at 10 t/day.

An annual floor lets the operator choose which days to preserve sales. Table 4 compares floors against the 215.1 TWh sold under A without investment. The 99.99% floor binds in this case.

Table 4. Annual sales floors in A. The first row shows unrestricted sales. The lost sales column reports realised reductions.

Scenario	Sales retained (%)	Lost sales (GWh)	H_2 (MW)	Gain (£m/yr)
Unrestricted	99.9266	157.92	20	8.57
Restore half the loss	99.9633	78.96	20	5.19
Tight sales floor	99.9900	21.51	20	1.04
Preserve all sales	100.0000	0.00	0	0.00

Model checks. The 2023 joint Xpress model is optimal (2,556 rows, 1,827 columns). Checks reconcile flows and revenues in 42 optimal and two rule schedules, with physical residuals below 0.001 MWh. Xpress confirms eight enumerated contract points spanning zero to expansion. Independent results match 18 cases [9].

2.6 Robustness, validation and returns

The 20 MW capacity is fixed in the 2022–25 replays, with dispatch reoptimised on each realised year. The site-only case gives background generation priority and limits device input to the simulated farm.

Table 5. Fixed 20 MW replays. Values are gains against the corresponding zero-investment case, in £m/year.

Period and assumptions	A. Economic dispatch	B. Daily electricity priority
2022	8.91	7.31
2023	8.57	−1.07
2024	11.95	−1.00
2025	13.17	4.48
2025, £4/kg and 5 t/day	−0.24	−0.77
2023, wind-site boundary	8.17	−1.07

Both policies lose money in the 2025 downside at £4/kg and 5 t/day. At 10 t/day, B's 2025 break-even falls to £3.13/kg as surplus rises. In 2023, A still gains £7.29m/year at 25% higher initial installation cost and an 8% real discount rate, with stack cost held fixed. The strict-quality 2023 sample retains the base capacities.

Storage check. A full usable discharge on every positive-price day earns at most £48.7k per module in 2023 even with free charging, below the £96.4k annual cost. In A with one battery and no electrolyser, replacing the annual balance with $s_t = s_{t-1} + \eta c_t - r_t/\eta$, $0 \le s_t \le E_B b$ and $s_0 = s_{|\mathcal{T}|} = 0$ cuts operating value from £13.3k to £4.3k. This is below the lower fixed-cost case of £11.1k/year. Intraday and ancillary value lie outside this daily case.

Cash flow. The 2023 A case receives £23.97m in hydrogen netback per year, forgoes £12.39m in power sales and spends £0.52m on fixed operations. After £2.49m/year for capital and stacks, annual gain is £8.57m. For £29.21m paid upfront, payback is 2.64 years simple and 2.90 years discounted. Repeating 2023 operations for 30 years with renewals in years 11 and 22 and zero residual value gives £131.78m net present value.

3 Tier 1 recommendation

Retain the existing assets while confirming hydrogen acceptance and delivered netback, site access to electricity and sales obligations. Consider one 20 MW module if the agreed terms clear the relevant threshold. Revisit storage when a paid service or a different operating window can cover its costs.

References

[1] MMCS. *2026/27 Project Brief*, pp. 2–3 and 5–6.

[2] MMCS Group 15. *Project data repair, release 20261104-v2*. Data, methods and audit.

[3] Office for National Statistics. Consumer Prices Index, D7BT.

[4] Danish Energy Agency. Technology data for energy storage, worksheet 180 Lithium Ion Battery.

[5] International Energy Agency. *Global Hydrogen Review 2025*, electrolyser investment costs.

[6] Department for Business, Energy and Industrial Strategy. *Hydrogen Production Costs 2021* and technical assumptions annex.

[7] Department for Energy Security and Net Zero. *Hydrogen Allocation Round 1 successful projects*, 2023.

[8] MMCS Group 15. *Financial conversions and technology cost basis*, R4, 2026.

[9] Clever User. *Independent Tier 1 implementation*, 20261105-v1, commit `024edd1`.